

Application of Design Thinking as a Teaching Method in the Industrial Design Course

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Abstract

This article has presented a model of teaching methods for the Industrial Design course with the application of "Design Thinking" and technology software such as Padlet, MS Form, and Quizizz. We tested and evaluated this teaching method model in the semester 2021.2 at Hanoi University of Science and Technology (HUST), Vietnam. The research results show that the indicators of the BLOOM scale of the new teaching method have increased by more than 10%, compared to the traditional teaching method. Especially in the groups of indicators of recognition, understanding, and analysis of art design elements increased by 18%. This teaching method has helped learners to develop comprehensively with "hard skills" and "soft skills". The teamwork method has also been applied, thus making the learning process more sharing and exchanging their varied knowledge, ideas, and opinions, which has contributed to broadening their understanding of topics. In addition, learners are equipped with critical thinking, analytical thinking, judgment skills, self-awareness skills, and innovative and creative spirit.

Keywords: Design thinking, Industrial design, teaching methodology.

1. Introduction

Modern society with the advent of industrial civilization has been developing at a rapid rate that has changed every aspect of social life. Therefore, the development model of education should approach the future. One of these is the most basic knowledge of design for industrial products. Therefore, Hanoi University of Science and Technology (HUST) has added the Industrial Design (ID) module to course elective units for all students since 2019. Over four years of teaching, we have researched and designed teaching methods for this course for engineering students. In this study, the authors applied "Design Thinking" (DT) [1] and Technology software in the teaching process. In the last several decades, DT has been increasingly popular in academic institutions. The ability to think creatively and collaboratively, as well as the ability to communicate effectively, can all be enhanced by regular practice of DT [2-9]. Accordingly, the contributions of this study to engineering students are identified [2]:

- Ability to analyze and evaluate the aesthetic aspect of the product
- Ability to illustrate problems in language and images using information technology tools
- Ability to promote self-learner, creative thinking, and teamwork skills.

The remainder of this paper is organized as follows. Section 2 describes the research methodology. Section

3 presents the experiment. Results and discussion are presented in section 4. Finally, section 5 presents the paper's conclusions.

2. Methodology

This research is carried out according to the basic tenets of education; learner-centered and DT methodology. ID classes provide a framework for students to engage with new learning by making connections, thinking critically, and exploring possibilities [2,3].

Teaching is a process by which the teacher brings the student and the subject matter together. It is not only telling and testing, but it is the complex art of guiding students through a variety of experiences and activities toward the attainment of goals. The basic lecture is a method of teaching by "chalk and talk". However, with the new age of technology and social media, lectures are the modern method of teaching that gives students the ability to acquire more knowledge through diverse means. This method has proved to be more efficient and productive since it employs the use of the internet, video presentations, games, etc which aid students' interaction and improves the entire learning process. As well as working to improve skills at lecturing, the instructor might also determine if the lecture approach is the best method of teaching. For effective teaching to take place, a good method must be adopted by a teacher. A variety of strategies and methods are used to ensure that all students have equal

opportunities to learn. A lesson plan may be carried out in several ways: questioning, explaining, modeling, collaborating, and demonstrating.

In 2010, Old Dominion University introduced learner-centered teaching method. This method is focused on the learners' needs, abilities, interests, and learning styles with the teacher as a facilitator of learning. Through research, experimentation, and practice activities, this method develops learners' critical thinking, and emotional intelligence and encourages learners' inquisitiveness and curiosity. The aim is to make a dramatic change in the teaching and learning process, as the shifts focus from acquiring knowledge (concepts and skills) to be able to apply it in a particular context with appropriate cognitive

strategies. This method needs to change the proportion of theory and practice hours by providing real-life situations, which applied knowledge and experience, and change the role of mere teachers into real mentors [4].

The DT method [1] was applied in this study by applying the process of design thinking to rebuild detailed and specific tasks for both the instructor and the learner. Specific tasks of lecturers and students through five general steps in the DT process are presented in Table 1 [4,5].

Table 1. Application of Design Thinking in teaching and learning

Steps of Design Thinking	Teacher's task	Student's task
Empathy 	- Determine who your students are? - What is their aspiration to study ID ? - What do they already know about the factors of ID ?	- What course should be studied? - Characteristics of teachers going to class - Methods of learning theory and exercises?
Determine 	- Student's needs and preferences? - Teaching methods? - Probe, ask questions and get feedback on design lessons.	- What should we study this subject for? - How to study effectively? - What is the main content? - Which chapter is the main - What skills are acquired?
Idea 	- Ask questions like how can students absorb this knowledge well? - Can write down ideas associated with themes, associated with visual illustration	- Find out how to learn wisely - How to understand and apply this knowledge
Prototype 	- Teachers can build activities to develop new ideas - Get Feedback from students. Evaluate, process information, and adjust other lesson designs.	- Make a study plan - Develop a suitable learning method for yourself and the subject (theory and practice part) - Build a study space - Determine the golden hour to study...
Test 	- Assess the ability to acquire knowledge - Evaluating teaching effectiveness - Interaction between teachers and students	- Evaluating learning effectiveness - Ability to apply learned knowledge - Practical experience

3. Experiment

ID [6,7] is the design activity of prototyping industrial products, beginning with ideas from clients and finishing with product design. After determining the roles of lecturers and students according to the steps of the DT process (Table 1), the task of “Research on the design of teaching methods for the ID course” includes:

- Research characteristics of students in HUST. This is the step of empathizing with learners in the design process.
- Research and design the content of the lectures and exercises. These are the Determine step and Idea Step.
- Evaluating the lecture will be the content of the Prototype step and test sample evaluation.

3.1 Research on student characteristics of HUST

HUST has many students from different majors such as Information Technology, Mechanical Engineering, Electricity, Technical Physics, Environmental Science, ect. So these students' knowledge of arts will be limited. The step of empathizing shows the students in HUST are still very shy with this course because they are afraid, they are not able to draw anything well. However, today, product styling is a very important issue in product development. Therefore, the analysis of the characteristics of HUST students indicates that the design of the lecture content and teaching demo ID lessons is a necessary mission.

3.2 Designing lesson content

Instructors want students to learn more about the concept of art and the related knowledge. As a result, learners know about products and aesthetic design from the knowledge they have learned. For making lectures for the ID module, it is necessary to answer the questions [2,4,5]:

- + What learners will acknowledge ?
- +How instructors will introduce and teach this module ?
- +What students will have to do and why ?



Fig.1 The idea of designing lessons for ID

The idea is given to build a teaching method to help HUST students perceive the elements of art from easy to difficult, from simple to complex. This teaching

method is learner-centered, and the teacher is considered an instructor during the lesson [4,7]. The model of the lesson we have designed is presented as shown in Figure 1.

The ID module needs visual illustration, beautiful color images, and the update of the information on science and technology, so we have more ideas about giving software support to the lecture workers. Padlet software is used to create an online group workspace that allows teachers to monitor the progress, and quality of assignments and promote the ability of each student. The system of exercises displayed on the software gives students more competition in learning. Another application that has been exploited is the “Quizziz” software. It is used to quickly test the knowledge students have learned through minigames. “MSform” software is also used to check or get feedback from learners about the quality of teaching. To create climaxes in learning, “Wheelofnames” software has been used to randomly select people to answer situational questions.

3.3 Lecture experiment

A general cycle for the lecture on Industrial Design taught to engineering students has been studied and presented as shown in Fig. 2.



Fig.2. A cycle for the ID lecture

In the cycle of Fig. 2, the steps are analyzed in more detail as follows:

- The lecturer introduces students to the concepts of art, art design, its expressive value, and applicability by PPT presentation combined with chalk and board.
- Through the look of facial expressions and eyes, the teachers should empathize with learners. By asking questions, the teachers understand the problem students are facing, thus arousing the curiosity, which will try to learn and be more aware of this problem.
- + To gain a deeper understanding of the concepts and processes of the art design, students are assigned to perform specific exercises on “Padlet” software (Fig.3).

On the “Padlet” pages of each group's exercises, the lecturer actively interacts directly with each student to

give feedback or give appropriate suggestions for the exercises.

+ After collecting all the information, learners conduct analysis, synthesis, or evaluation of the assigned task in the exercise. Each group of students will present a summary of the work in front of the class. They will listen and absorb ideas from other students as well as lecturers to improve their assignments.

+ At the end of each chapter, the lecturer will give questionnaires to systematize knowledge for learners about that chapter.

Thus, the steps mentioned above have formed a general cycle for the teaching method of the course ID.

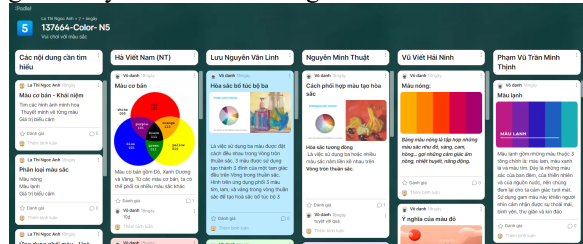


Fig. 3. Deploy the exercise to learn about art design on “Padlet” software

4. Result and Discussion

4.1 Study on characteristics of students from HUST

The results of a survey of 168 students from different majors showed that they actively inform themselves about the ID course via the university’s website as well as listening to feedback from past participants of the course. This is illustrated through the graph in Fig. 4 and 5.

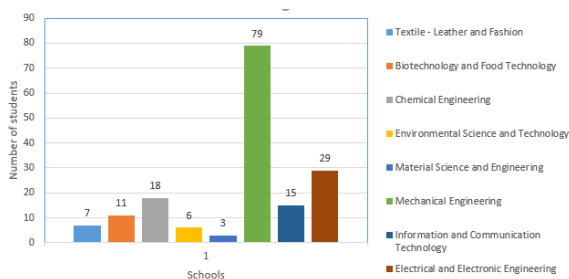


Fig. 4. Survey results of faculties and institutes with students registered to study the ID course.

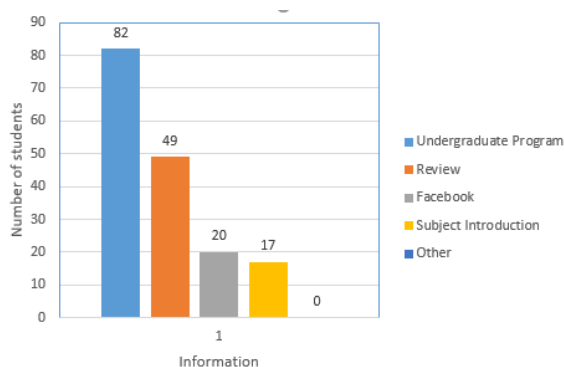


Fig. 5. Survey results on student information sources about the ID course.

HUST’s students often choose to register and study the ID course from the 3rd year onwards (Fig. 6). After being equipped with the basic knowledge of the specialization they realized that this subject is important because of its role in the design of products.

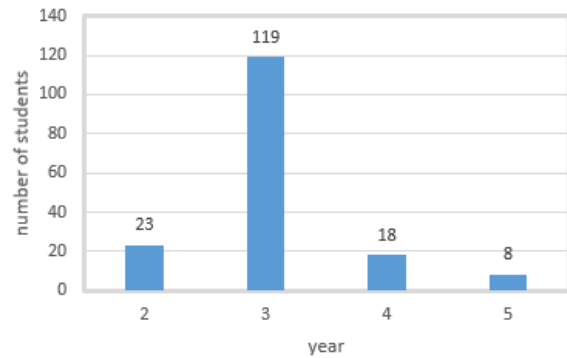


Fig. 6. Results of the survey of what student year they enrolled

When it comes to ID, HUST’s students still have limited knowledge of 5 elements of Art, especially color and composition in the design (Fig. 7). Therefore, we have proactively proposed teaching methods focused on this issue to help students gain an improved knowledge of industrial product design in subjects, requiring elements and principles of art in design.

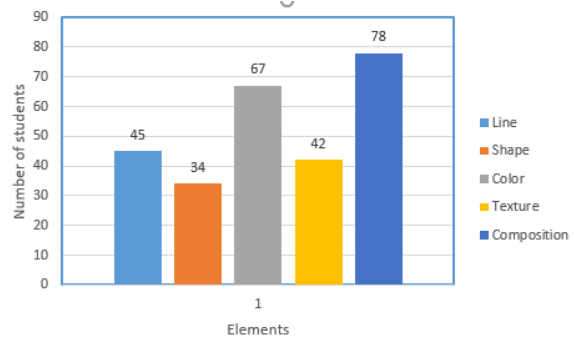


Fig. 7. Survey results on the limited content of knowledge about ID of HUST’s students.

4.2 Research results on the application of new teaching methods for the ID course

The research results assess the change in teaching quality between the two semesters: 2020.1 (this is the second semester of this module and is taught by the traditional method) and the term 2021.2 (the new method has applicated modern technology, after being adjusted through the periods 2020.2 and 2021.1) is

presented on the chart in Fig. 8 according to the criteria of the BLOOM rating scale [4].

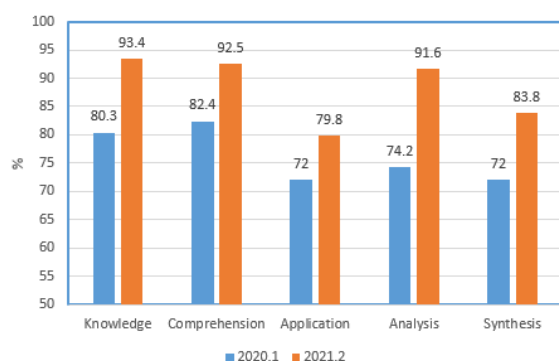


Fig. 8. Results of evaluation of teaching methods according to the BLOOM scale

5. Conclusion

The research has applied the method of "design thinking" in building the lecture and teaching the Industrial Design module to HUST students. In addition, this study evaluated this lecture model. In combination with the use of software such as Padlet, MS Form, and Quizizz in teaching, learners have gained more knowledge of Fine Arts. The evaluation of the BLOOM scale for the new teaching methodology is higher than the traditional teaching methodology by over 10%. Moreover, we also see real progress in the perception of art elements, and analysis of these elements, which has increased by almost 18%. However, the ability to apply the knowledge learned about product design is not high, due to the lack of experience in design practice. This is also recorded as the next goal that the teachers want to achieve. In addition, learners are equipped with critical thinking, analytical thinking, judgment skills, self-awareness skills, and the spirit of innovation and creativity.

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